

Adaptive Patience: Learning When Proactive AI Should Offer Help in Spatial Search Tasks

ANONYMOUS AUTHOR(S)

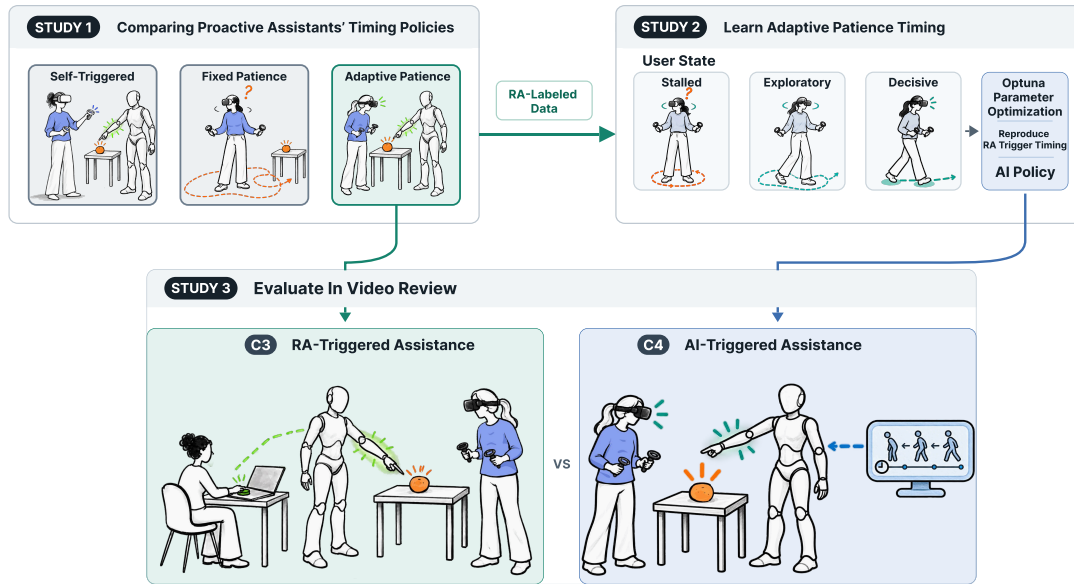


Fig. 1. Overview of Studies 1, 2, and 3. Study 1 compared self-triggered, fixed-patience proactive, and research-assistant-triggered adaptive assistance. Its behavioral traces and interventions served as reference data for calibrating the automated policy in Study 2. Study 3 compared the research-assistant-triggered and AI-triggered adaptive implementations during live use and retrospective video review.

Proactive AI must balance independent action with assistance needs. We introduce *Adaptive Patience*, an interpretable timing policy mapping observable behavior to state-specific waiting thresholds. Across three VR kitchen-search studies, we evaluated the policy, its automation, and live deployment. In a within-subject Wizard-of-Oz study ($N = 69$), Adaptive Patience showed patterns: relative to Fixed Patience, it was less disruptive, less wrong-timed, and more partner-like; relative to Self-Triggered assistance, it was more partner-like but more disruptive and less controllable. Using behavioral traces, we automated Adaptive Patience by calibrating a policy-constrained state machine with Optuna, achieving .847 user-independent balanced accuracy. In a live within-subject study ($N = 15$), we deployed AI- and RA-triggered assistance. Interviews revealed that participants judged timing by the remaining value of independent search, varied in their desired patience, and weighed reduced help-seeking burden against autonomy. We contribute an interpretable framework, an automated implementation, and design guidance for timely, autonomy-preserving proactive AI.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Interactive systems and tools**.

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1 Introduction

AI assistants are increasingly designed to offer support without waiting for an explicit user request. Such proactive systems have been developed for programming, memory support, and situated physical activities such as assembly and mixed-reality guidance [17, 32, 38, 39]. Once a system can initiate assistance, however, its helpfulness depends not only on what it offers, but also on *when* it offers it. Assistance that arrives too early can interrupt productive, independent activity, whereas assistance that arrives too late can withhold support until much of its value has passed. Prior work therefore characterizes initiative timing as a balance between the cost of interrupting ongoing activity and the cost of delaying potentially useful support [23]. These interruption costs also vary across tasks and contexts [1, 38]. Good proactive timing is not simply acting as early as possible: it should preserve opportunities for independent action while offering support when users experience it as appropriate, satisfying, and useful rather than disruptive [38].

Existing approaches offer some solutions for timing proactive assistance, but each involves tradeoffs. One common strategy is fixed-time assistance, which is offered after a preset amount of task-independent progress time has elapsed [38, 48]. Another relies on known task structure, offering help at task boundaries—for example, after a user completes one subtask and begins formulating the next [32, 38]. These strategies are straightforward and predictable, but they do not adapt the system’s timing to whether a user is making meaningful progress, productively exploring, or stalled.

Other systems instead use sensed user behavior or task-progress cues to decide whether to offer help immediately [8, 22, 39, 44–46]. For example, a VR wayfinding system used users’ gaze patterns to predict their need for navigation aid and triggered a map or directional arrows accordingly [8]. This approach is valuable, but it leaves an important timing challenge in open-ended tasks such as spatial search. For example, scanning behavior can reflect either that they are stuck or that they are actively exploring, depending on the broader task-related progress [33, 45]. Moreover, the need for help often develops through persistent difficulty and an accumulated lack of progress, rather than through a single instantaneous behavior [4, 39]. Therefore, instead of treating one observed behavior as an immediate trigger for help, a timing policy needs to interpret behavioral signals as evidence of how a task is unfolding and use that interpretation to determine how long to leave room for independent action.

We introduce **adaptive patience**, an interpretable proactive-assistance timing policy that makes this temporal link between user state and assistance explicit. The policy combines behavioral and contextual evidence to infer the user’s state. Each state selects a patience threshold: the amount of independent action time reserved before assistance becomes eligible. States with higher assistance needs lead to shorter patience, while more on-course, decisive action leads to longer patience (See Section 3). Consequently, exploratory behavior early in a task can proceed without interruption, whereas the same behavior later in a task can make assistance eligible if it persists.

We instantiated adaptive patience in an embodied spatial-search task for two reasons. First, these tasks are challenging because users must continually decide where to move, what to inspect, and which objects to pursue while the target’s location is unknown [2, 8]. Second, assistance timing matters for this task because early intervention can interrupt users’ on-course progress, while late intervention can cause confusion, frustration, and poor performance [1, 23]. We examined the policy in a virtual-reality kitchen, where participants searched for a sequence of target ingredients under

an overall time constraint. Although we evaluate adaptive patience in spatial search, the principle may also inform game-hint and intelligent-tutoring systems, where assistance can directly advance a subgoal but should be deferred while independent exploration or problem solving remains productive.

Within this task, we posed the following research questions:

RQ1: In embodied VR spatial search, how does adaptive proactive assistance compare with self-triggered assistance and fixed-time proactive assistance in terms of task performance and user experience, including satisfaction, partner role, and control?

RQ2: How closely can an automated implementation reproduce whether and when trained research assistants (RAs), applying the adaptive-patience policy, triggered assistance?

RQ3: During live deployment, how does AI-executed adaptive patience compare with RA-executed adaptive patience in system usability, user experience, and participants' interpretations of timing and control?

To answer these questions, we conducted three linked studies (Figure 1). Study 1 used a within-subject experiment to compare self-triggered, fixed-time proactive, and adaptive proactive assistance. In the adaptive condition, trained RAs executed the predefined policy through a Wizard-of-Oz procedure. The resulting behavioral traces and intervention timing served as references for automated policy in Study 2. Study 2 used adaptive-condition data to calibrate, via Optuna-based parameter optimization, a policy-constrained state machine that reproduced RA intervention decisions and timing based on observable signals. We evaluated decision and timing on both user-dependent and user-independent validation protocols. Study 3 deployed the automated policy and compared AI-executed with RA-executed adaptive patience during the same tasks as Study 1, and interviewed participants about their subsequent review of each assistance event. Together, these studies trace adaptive patience from policy design and controlled comparison, through behavioral calibration, to real-time deployment.

Our work makes three contributions:

- (1) A design exploration of *adaptive patience* that enables proactive assistants to vary how long they wait before intervening according to users' observable states. This framing treats assistance timing as preserving a state-appropriate opportunity for independent action.
- (2) An end-to-end technology probe that instantiates adaptive patience in embodied VR spatial search, spanning trained research assistants' Wizard-of-Oz enactment of the policy, an Optuna-calibrated timing model, and real-time AI deployment.
- (3) A multi-study empirical investigation of adaptive patience across controlled comparison and live AI deployment. The findings characterize its implications for task performance, perceived timing, and user agency, and identify design implications for future proactive assistants.

2 Related Work

2.1 Timing Policies for Proactive Assistance

We define a *timing policy* as the set of rules through which a system determines when to intervene proactively [23, 41, 47]. The timing of proactive assistance involves balancing the expected benefit of acting against the cost of interrupting ongoing activity and the cost of deferring useful support [23]. Research on interruptions further shows that, even when the information provided is identical, its timing can affect task performance, subjective experience, and users' evaluations of system behavior [1, 12, 25]. Timing is thus a core design dimension of proactive assistance [21, 34].

Prior systems have typically timed interventions using duration thresholds and recognizable task events [10, 11, 38, 40, 41]. For example, assistance can be triggered using duration-based rules, under which assistance becomes eligible only after a pre-specified amount of total time, idle time, cooldown time, or maximum waiting time. In programming, Chen and colleagues use activity-gating rules in an editor: suggestions are shown only after the user has stopped typing for 5 s and when at least 20 s have elapsed since the previous suggestion, interaction, or chat; in contrast, a code-execution error triggers an immediate debugging suggestion [17]. Similarly, some such systems wait for a more appropriate interruption point while imposing a preset maximum deferral window; if no better boundary appears, they still intervene once that deadline is reached [24, 26].

Some timing policies link interventions to recognizable events in a workflow. CodeCollaborator treats the completion of a code block, program execution, and pasting multiple lines of code as opportunities to offer checking, explanation, optimization, or debugging support. It also combines these events with temporal rules, such as 30 s of inactivity and 15 s of continuous text selection [38]. Such recognizable events provide clear signals of users' needs for help, reducing the risk of interrupting them at an arbitrary point in their work [12, 26].

Prior systems have used waiting mechanisms or task-event information. However, fixed patience cannot adapt the timing threshold between productive independent action and exploratory or stalled states, which are usually associated with a higher need for help. The following sections explain why adaptivity matters in embodied spatial tasks and why help seeking provides a theoretical basis for an adaptive patience timing policy.

2.2 Proactive Assistance in Embodied Spatial Tasks

Embodied spatial tasks provide an important yet challenging setting for proactive assistance [8, 32, 44, 46]. To locate a target object, navigate to a designated location, or perform a physical procedure, users must continually decide where to go and how to advance their current goal based on information from their surroundings. AR and VR systems can capture locomotion, head movement, hand movement, and gaze to help researchers understand user state [19, 46, 51], while placing spatial assistance like an agent or a map directly in the space [29, 30, 32].

Prior work demonstrates that real-time behavioral and environmental context can be used to determine whether and what spatial assistance to present [8, 9, 44, 46, 51]. "Lost in Style" used sequences of gaze direction and head orientation to predict whether users would need navigational support in a VR city; when the prediction was positive, the system displayed an arrow or mini-map [8]. Truong-Allié and colleagues designed a system to offer a navigation arrow when detecting visual search from head, hand, and gaze motion [46]. Similarly, Zhao and Cheng used eye-tracking features to predict whether users exploring a VR environment needed navigation assistance and to drive real-time adaptive navigation [51]. More recently, in a real AR spatial search-and-pick task, Seeliger and colleagues combined task progress, gaze, head movement, environmental depth, and time since the previous cue to predict whether users currently needed a navigation cue. The system then highlighted the target once the predicted need was positive [44]. Together, these studies show that embodied systems can use behavioral, environmental, and task information to align assistance with users' current activities or known workflows [29, 49]. In summary, many spatial assistance systems still follow a timing logic of inferring a current state and then immediately showing or hiding a cue [8, 9, 44, 46, 51].

In open-ended embodied search, states associated with the need for help do not always signal a stop in individual exploration and an immediate need for assistance. Scanning the environment, briefly pausing, changing direction, or adjusting movement speed can occur when users are having difficulty, but they can also reflect active exploration, narrowing of the search space, or progress toward a target [2, 20, 33, 52]. Current observable behavior and the resulting user state could therefore serve as evidence of how a search is unfolding, and users may need assistance in the near

Table 1. Mapping the three Adaptive Patience design requirements to the general policy and its spatial-search instantiation.

Design requirement	General policy commitment	Spatial-search operationalization
R1: State-conditioned waiting	The current user state selects a patience threshold of how long the assistance should wait. A state associated with lower help needs leads to longer patience of independent time.	When the On-course progress gate is inactive, Decisive, Exploratory, and Stalled states select thresholds of 15, 10, and 5 seconds, respectively.
R2: Persistent temporal context	An episode clock accumulates throughout the task. A state change updates the applicable threshold without resetting elapsed time.	The search clock begins when the trial starts and continues across movement, scanning, pausing, and reorientation.
R3: Progress protection	Meaningful observable progress toward the goal protects continued independent action.	Meaningful movement toward the known target activates the On-course progress gate, withholding assistance while that progress remains current.

future if the task is not progressing well [18, 33, 45]. Existing spatial systems establish the feasibility of state-sensitive assistance, but few explicitly specify how a user's state should determine how long to preserve independent exploration before proactive assistance becomes eligible.

2.3 Help Seeking as a Temporally Extended Decision Process

Research on help seeking suggests that requesting assistance is not an automatic response to the first sign of difficulty, but a self-regulated decision that unfolds over the course of problem solving [7, 36, 37]. Theoretical accounts distinguish several stages: users first recognize a potential need for help, monitor their task progress, may continue independent attempts in their current state, and eventually decide whether to seek assistance [5, 7]. Throughout this process, users evaluate whether further independent action is likely to yield progress [37, 42]. Independent problem solving typically precedes help seeking, and help ideally supports a return to independent problem solving [7, 37]. Thus, observable difficulty, search behavior, or limited progress are important inputs to a help-seeking decision, but leaving patience for users' agency in independent action before providing help remains important [4, 6].

Moreover, empirical work further suggests that observable difficulty does not map one-to-one onto immediate help seeking [4, 6, 33, 42, 43]. Whether and when users request assistance can also be shaped by their desire to retain autonomy, concerns about competence or social evaluation, embarrassment, and concerns about becoming dependent on others [14, 16, 27, 28, 43]. A period of continued independent action after a user exhibits difficulty may therefore provide valuable space to assess the environment, develop a strategy, or verify a next step [6, 37, 42].

Therefore, timing for proactive assistance in embodied spatial search should account for both user state and time: the system should preserve a state-appropriate period for independent exploration and offer help when behavior indicating a need for assistance persists without meaningful target-directed progress. We therefore design *adaptive patience* as a user-state-sensitive waiting policy.

3 Adaptive Patience: A General Timing Policy and Its Spatial-Search Instantiation

We propose *Adaptive Patience*, a timing policy for deciding how long a proactive system should defer direct assistance during a goal-directed task. The policy uses the observed *user state* to select a state-appropriate allowance for independent action and compares that allowance with persistent elapsed task time. Table 1 summarizes the three design commitments

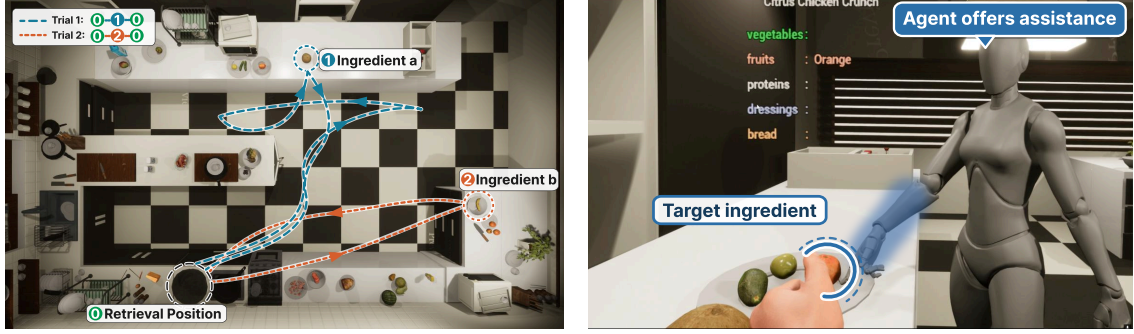


Fig. 2. The embodied ingredient-search task and standardized assistance. Left: the virtual kitchen and ingredient locations. Right: the assistant appears near the target, points toward it, and verbally indicates its location.

and their spatial-search instantiation. We first define the general mechanism and then describe its task-specific operationalization.

3.1 General Adaptive Patience Policy

At each decision point t , the system estimates the current user state u_t from observable behavior. Here, user state is an interpretable characterization of how a person is currently engaging with the task—for example, pursuing a coherent strategy, exploring alternatives, or appearing stalled. It is not, by itself, a judgment that the person needs help. The policy maps u_t to a *patience threshold*, $\tau(u_t)$, which specifies how much elapsed task time the system should reserve for independent action while that state is current.

Let T_t denote the time elapsed since the task trial began. While the trial remains incomplete, the system continues waiting until T_t reaches the patience threshold associated with the current user state, at which point it offers assistance. A change in user state updates the applicable threshold but does not reset T_t , which prevents the waiting period from restarting whenever behavioral state changes. Thus, user state determines the amount of independent time currently sought, whereas elapsed time records the opportunity already provided.

3.2 Instantiation in Embodied Spatial Search

We instantiated Adaptive Patience in an immersive virtual-reality ingredient-search task in Unreal Engine 5.4. The task comprised multiple discrete search trials with open-ended routes, objectively defined targets, and standardized assistance. These properties allowed us to vary when assistance was initiated while controlling the task settings.

Trial and assistance setting. Within each condition, participants completed up to nine target-search trials within a three-minute time limit. Before each trial, they were shown a target ingredient and then navigated the virtual kitchen to locate it, grasped it by pressing the trigger button on either controller, and carried it to the designated retrieval position. To exclude the time participants spent viewing the target prompt, the trial clock began at active-search onset, defined as the first sample after target presentation at which translational speed exceeded 50 cm/s. Delivering the ingredient to the retrieval position completed the trial and initiated the next target search.

Ingredients were distributed throughout the kitchen (Figure 2, left), including inside baskets and a cabinet. The nine-target sequence and ingredient–location map were randomized, and no participant encountered the same sequence–map combination twice.

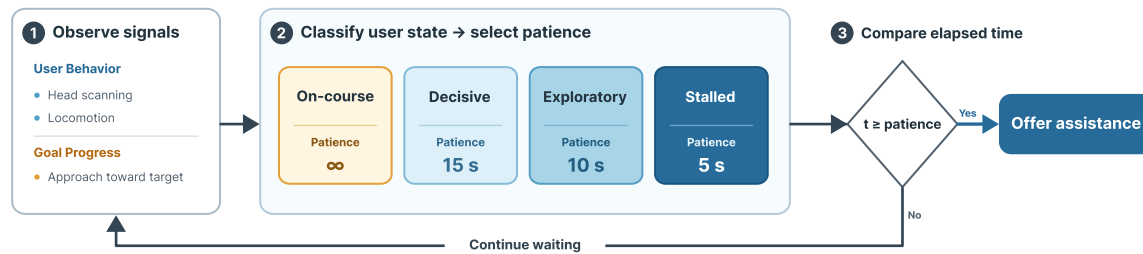


Fig. 3. Adaptive Patience and its spatial-search instantiation. Observable behavior informs the current user state, which selects a patience threshold. One persistent trial clock is compared with the threshold and help is offered when the patience threshold is reached.

When assistance was initiated, the virtual assistant appeared near the target, pointed toward it, and verbally said "it is here" (Figure 2, right). Its appearance, guidance content, and delivery were constant across conditions; only the timing policy varied. We used instantaneous appearance rather than movement from a fixed location to prevent delivery latency from confounding assistance timing or task performance.

User states from observable signals. The spatial-search rubric was informed by prior characterizations of search and wayfinding behavior [2, 52]. Movement clearly toward the target activated the ON-COURSE state. When this state was inactive, DECISIVE described sustained movement along a coherent route; EXPLORATORY described active scanning, reorientation, or movement among alternative directions; and STALLED described continued visual search with little or no translational movement.

User-state-to-patience mapping and pilot calibration. ON-COURSE is implemented as a progress-protection state with the longest, infinite waiting patience. DECISIVE movement receives long, 15 s patience, active EXPLORATION gets a shorter, 10 s patience, and STALLED with little spatial advancement gets the shortest, 5 s patience (Figure 3).

We selected the task-specific parameters through pilot testing with five participants before Study 1. We anchored the longest finite threshold to the mean latency to a help request in the pilot's self-triggered condition, and set shorter thresholds to provide progressively less time for states showing weaker evidence of continued advancement, while about half of pilot trials activated the on-course state. The same pilot mean set the fixed-time threshold used in Study 1's Fixed Patience condition. We selected the mapping to avoid floor- or ceiling-level intervention rates and to keep expected assistance rates broadly comparable across conditions, so the comparison primarily reflected differences in assistance timing rather than assistance frequency.

4 Study 1: Wizard-of-Oz Evaluation

Study 1 tested whether Adaptive Patience changed task performance and user experience relative to user-controlled and fixed-time assistance (RQ1).

4.1 Participants

We recruited 69 participants (46 females, 23 males, and 0 other), aged 18–46 years ($M = 22.55$, $SD = 4.64$), through e-mail solicitation to students and faculty at a mid-sized university in the northeastern United States. Participants identified as Bi-racial, mixed, multicultural, or multi-racial ($n = 6$); Black or African heritage ($n = 5$); Central Asian ($n = 1$); East Asian ($n = 28$); Hispanic, Latino, Chicano, or Puerto Rican ($n = 9$); Middle Eastern ($n = 1$); South Asian ($n = 9$);

Southeast Asian ($n = 5$); and White or of European ancestry ($n = 16$). Because multiple selections were permitted, these counts are not mutually exclusive. Each study session lasted around 45 minutes and participants were compensated with \$15. The study was approved by the institutional review board at [redacted for peer review].

4.2 Conditions and Procedure

We used a within-subjects design with three counterbalanced conditions: *Self-Triggered* (C1), *Fixed Patience* (C2), and *Adaptive Patience* (C3). Self-Triggered provided a user-controlled reference for assistance timing. Fixed Patience provided a transparent proactive baseline that waited for a single, state-independent duration, but did not condition the waiting time on observed user state. The virtual environment, search task, and visual and verbal assistance behavior described in Section 3.2 were held constant across conditions; only the policy that initiated assistance varied.

- **Self-Triggered (C1).** Participants could request assistance at any time by pressing the A button on the right-hand controller.
- **Fixed Patience (C2).** The assistant initiated assistance automatically 15 s after trial start if the target had not yet been retrieved.
- **Adaptive Patience (C3).** Two trained research assistants (RAs) independently monitored participants' state and jointly enacted the adaptive-patience policy described in Section 3. They made an initial state assessment after the first 3 s of each search, and reassessed it at 3-s intervals thereafter. At each assessment, an RA classified the participant's current state as *On-course*, *Decisive*, *Exploratory*, or *Stalled*, and gave the user infinite, 15 s, 10 s, or 5 s patience for independent search accordingly.

The RAs then compared the elapsed time displayed by the trial clock with the patience threshold associated with the current state. Each RA signaled that assistance was eligible with a thumbs-up action only when the patience threshold had elapsed. When both RAs had signaled, a designated RA then issued the keyboard command to trigger the agent offering assistance.

After providing consent, participants completed training on locomotion, object manipulation, and interaction with the assistant. They then completed three 3-minute condition blocks, each containing up to nine sequential ingredient-search trials, in random order across conditions C1 to C3. After each block, participants completed the subjective measures in the post-condition survey described below.

4.3 Measures and Analysis

Measures. We first calculated *assistance counts* as a process measure to report how often each policy intervened. This includes the number of assistance events per participant and the proportion of trials containing at least one event. *Objective performance* comprised completion rate, completion time for completed trials, path tortuosity (fractal dimension), and wrong-grab count, capturing task success, efficiency, path complexity, and interaction errors.

Subjective measures in the post-condition survey comprised NASA-TLX, SUS, and ten five-point items (1 = *Strongly disagree*; 5 = *Strongly agree*) assessing effort (Q1), partner perception (Q2), mutual awareness (Q3–Q4), control (Q5), mutual understanding (Q6–Q7), disruption (Q8), intervention at the wrong time (Q9), and timing satisfaction (Q10) [38, 50]. NASA-TLX and SUS were scored as multi-item aggregates. Higher Q1, Q8, and Q9 scores were less favorable; higher scores on the other items were more favorable. Figure 4 provides the full item wording.

Analysis. Condition effects were tested with Friedman tests, with Kendall's W as the omnibus effect size. We controlled the false discovery rate separately across the four objective outcomes and the 12 post-condition outcomes

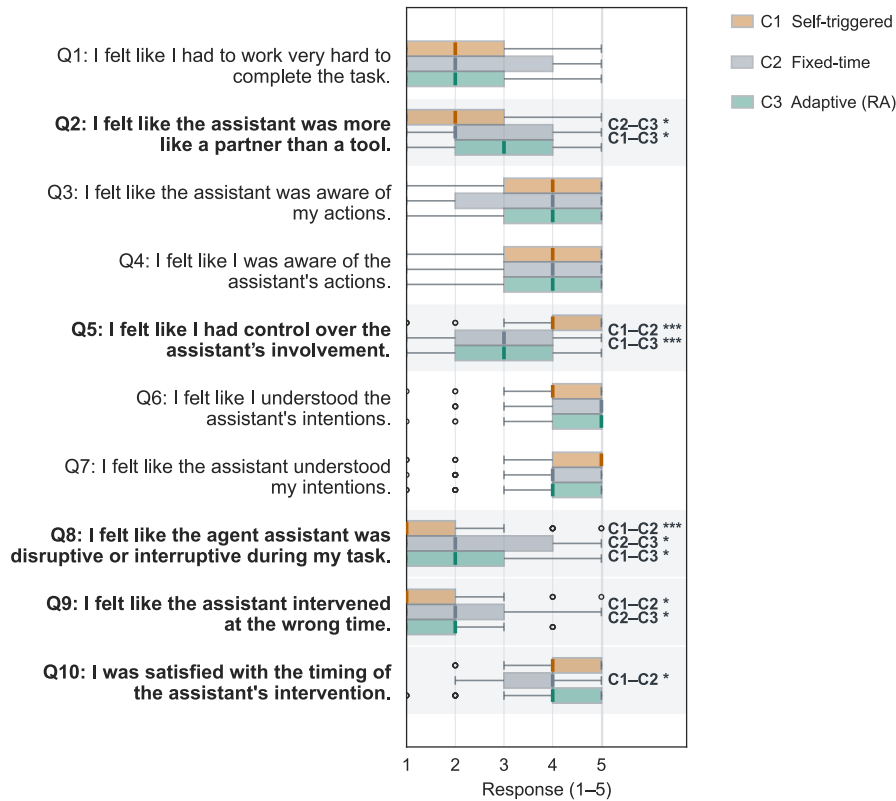


Fig. 4. Post-condition ratings under Self-Triggered, Fixed Patience, and Adaptive Patience. Boxes show interquartile ranges; colored lines show medians; whiskers extend to 1.5 times the interquartile range; circles denote outliers. Bold rows show BH-FDR-significant Friedman tests. Pair labels show Holm-adjusted Wilcoxon tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

using Benjamini–Hochberg correction. FDR-significant outcomes were followed by three pairwise Wilcoxon signed-rank tests with Holm correction; matched-pairs rank-biserial correlations (r_{tb}) quantified pairwise effects. All 69 participants provided complete condition-level data for the 12 post-condition outcomes. All tests were two-sided with $\alpha = 0.05$.

4.4 Results

4.4.1 Data Inclusion and Assistance Counts. The dataset contained 1,647 initiated trials from all 69 participants, including 71 incomplete trials. Assistance occurred in 587 trials (35.6%), with 606 events in total because 19 trials contained more than one event.

Fourteen participants had at least one condition in which no assistance occurred, but no participant received or requested no assistance across all three conditions. These participants were retained as specified above, and every logged assistance event was assigned to a trial.

Participants requested or received an average of 2.57 ($SD = 2.28$), 3.26 ($SD = 1.35$), and 2.96 ($SD = 1.47$) assistance events in Self-Triggered, Fixed Patience, and Adaptive Patience, respectively. At least one event occurred in 31.0% (168/542), 40.3% (221/549), and 35.6% (198/556) of trials, respectively. Event counts differed overall, $\chi^2_F(2) = 8.08$,

Kendall's $W = 0.059$, $p = 0.018$, but no pairwise contrast survived Holm correction: Self-Triggered versus Fixed Patience, $W = 515.00$, $p_{\text{Holm}} = 0.060$, $r_{\text{rb}} = -0.355$; Self-Triggered versus Adaptive Patience, $W = 674.00$, $p_{\text{Holm}} = 0.216$, $r_{\text{rb}} = -0.238$; and Fixed Patience versus Adaptive Patience, $W = 617.00$, $p_{\text{Holm}} = 0.274$, $r_{\text{rb}} = 0.169$.

Among trials containing assistance, mean latency to the first help event was 10.56 s ($SD = 11.63$), 14.79 s ($SD = 2.06$), minor deviations from the nominal 15-s threshold likely reflect timestamp alignment and frame-based event scheduling), and 15.07 s ($SD = 5.03$) in Self-Triggered, Fixed Patience, and Adaptive Patience, respectively. Trials without assistance were excluded from these descriptive timing summaries.

4.4.2 Objective Task Performance. Condition did not affect completion rate, completion time, path tortuosity, or wrong-grab count after BH-FDR correction (q range: 0.587–0.848). Omnibus effects were small (Kendall's $W = 0.002$ –0.026), providing no evidence that Adaptive Patience improved or impaired objective performance relative to either reference condition.

Trials in which the agent offered assistance were descriptively longer, more tortuous, and more error-prone than the full trial set, consistent with assistance being concentrated in more difficult searches. An exploratory analysis restricted to completed trials containing assistance nevertheless found no condition differences in completion time, path tortuosity, or wrong-grab count after correction (q range: 0.132–0.251; Kendall's $W = 0.025$ –0.061). Because assistance occurrence depended on how each search unfolded, this subset analysis is descriptive.

4.4.3 Subjective Experience. Five outcomes showed FDR-significant condition effects: partner perception (Q2), $\chi_F^2(2) = 8.96$, $W = 0.065$, $q = 0.027$; control (Q5), $\chi_F^2(2) = 17.10$, $W = 0.124$, $q = 0.002$; disruption (Q8), $\chi_F^2(2) = 16.34$, $W = 0.118$, $q = 0.002$; intervention at the wrong time (Q9), $\chi_F^2(2) = 9.12$, $W = 0.066$, $q = 0.027$; and timing satisfaction (Q10), $\chi_F^2(2) = 10.71$, $W = 0.078$, $q = 0.019$.

Relative to Fixed Patience, Adaptive Patience was perceived more as a partner ($M = 2.86$ vs. 2.59, $W = 124$, $p_{\text{Holm}} = 0.042$, $|r_{\text{rb}}| = 0.467$), as less disruptive ($M = 2.10$ vs. 2.38, $W = 135.5$, $p_{\text{Holm}} = 0.049$, $|r_{\text{rb}}| = 0.417$), and as less likely to intervene at the wrong time ($M = 1.86$ vs. 2.23, $W = 155$, $p_{\text{Holm}} = 0.025$, $|r_{\text{rb}}| = 0.479$). Timing satisfaction was descriptively higher under Adaptive Patience ($M = 4.07$ vs. 3.77), but the contrast showed a marginal improvement after Holm correction ($W = 208.5$, $p_{\text{Holm}} = 0.051$); perceived control also did not differ ($W = 449$, $p_{\text{Holm}} = 0.765$).

Relative to Self-Triggered assistance, Adaptive Patience was perceived more as a partner ($M = 2.86$ vs. 2.45, $W = 290$, $p_{\text{Holm}} = 0.042$, $|r_{\text{rb}}| = 0.414$), but also as more disruptive ($M = 2.10$ vs. 1.77, $W = 170$, $p_{\text{Holm}} = 0.049$, $|r_{\text{rb}}| = 0.429$) and less controllable ($M = 3.25$ vs. 4.07, $W = 160$, $p_{\text{Holm}} < .001$, $|r_{\text{rb}}| = 0.662$). The two conditions did not differ in wrong timing ($W = 305$, $p_{\text{Holm}} = 0.467$) or timing satisfaction ($W = 414.5$, $p_{\text{Holm}} = 0.832$).

Compared with Fixed Patience, Self-Triggered assistance was less disruptive ($M = 1.77$ vs. 2.38, $W = 101.5$, $p_{\text{Holm}} < .001$, $|r_{\text{rb}}| = 0.695$), less likely to occur at the wrong time ($M = 1.74$ vs. 2.23, $W = 228$, $p_{\text{Holm}} = 0.011$, $|r_{\text{rb}}| = 0.495$), more satisfying in its timing ($M = 4.12$ vs. 3.77, $W = 264.5$, $p_{\text{Holm}} = 0.044$, $|r_{\text{rb}}| = 0.414$), and more controllable ($M = 4.07$ vs. 3.29, $W = 137$, $p_{\text{Holm}} < .001$, $|r_{\text{rb}}| = 0.666$). Partner perception did not differ ($W = 394$, $p_{\text{Holm}} = 0.324$).

Condition did not affect NASA-TLX, $\chi_F^2(2) = 0.29$, $W = 0.002$, $q = 0.867$, or SUS, $\chi_F^2(2) = 3.58$, $W = 0.026$, $q = 0.250$. Effort, awareness, and understanding also showed no FDR-significant effects (all $q \geq 0.104$).

4.5 Study 1 Summary

Study 1 shows that Adaptive Patience improved the experience of proactive assistance relative to a behavior-insensitive fixed threshold. Adaptive Patience was perceived as more partner-like, less disruptive, and less likely to intervene at the wrong time, with descriptively larger timing satisfaction than Fixed Patience. These benefits were observed without

detectable pairwise differences in assistance frequency or differences in objective performance, locating the primary advantage of adaptation in the intervention experience rather than task efficiency.

Relative to Self-Triggered assistance, Adaptive Patience was perceived as more partner-like but also as more disruptive and less controllable, consistent with previously reported tradeoffs between proactive initiative and direct user control [38]. Meanwhile, timing-specific evaluations, including perceived wrong-time intervention and satisfaction with timing, did not differ detectably from user-triggered timing.

Additionally, unlike self-triggered requests, whose timing may reflect internal factors that are not observable to the system, Adaptive Patience provides a clearer basis for data-calibrated automation. Its explicit mapping from observable behavior to user state, patience thresholds, and intervention decisions yields an inspectable computational structure for learning from data. Accordingly, Study 2 used RAs’ consensus judgments about whether and when to intervene as reference outputs to calibrate an automated implementation of the policy.

5 Study 2: An Automated Implementation of Adaptive Patience

Study 1 evaluated adaptive patience through a Wizard-of-Oz implementation in which two trained research assistants (RAs) jointly applied the adaptive patience policy. Study 2 examined how closely an automated timing system could reproduce this implementation from continuously observed behavioral and environmental data, enabling the policy to operate without human observers in future proactive systems (RQ2).

5.1 Goal and Reference Data

Our goal was to reproduce when the adaptive-patience policy, as operationalized by the trained RAs in Study 1, issued assistance. From previous literature, a self-triggered request reflects a participant’s internal decision to seek help, which may be delayed, withheld, or shaped by factors not fully observable in behavior [16, 28]. In contrast, under Adaptive Patience, RAs applied a shared policy using externally observable behavioral signals. Their trigger times therefore provided time-stamped reference implementations of the policy that an automated system could learn to approximate.

We used data from the Adaptive Patience condition of Study 1 only. The recorded data included participants’ head and body movement and target-relative spatial information. Of 556 initiated trials from 69 participants, we excluded 15 incomplete trials and retained 541 trials for evaluation. These trials included 189 RA-triggered trials and 352 RA-no-trigger trials.

We note that the RA decisions are not error-free labels. Although the RAs used a predefined rubric and had to reach consensus before triggering assistance, real-time judgment could still involve ambiguity when user states changed. In addition, a brief delay could arise between recognizing that a patience threshold had been reached and issuing the keyboard command. Accordingly, Study 2 evaluates how closely the automated policy can reproduce the deployed RA implementation, rather than claiming that it identifies an objectively optimal intervention time.

5.2 Policy-Constrained Automated Implementation

In Study 1, the RAs applied *Adaptive Patience* to observable behavior and task progress. To automate the policy while retaining its explicit state-machine structure, we implemented a parameterized state-machine model that encoded its task-progress states, state-specific patience thresholds, and progress gate. At each update point, the model interpreted recent behavioral evidence through a weighted voting procedure to infer a task-progress state. The inferred state selected a patience threshold, which the system compared with continuously accumulated trial time to determine whether to offer assistance. We used Optuna-based parameter optimization [3] to calibrate the model’s temporal

representation and state-inference parameters so that its trigger decisions and trigger times approximated the RA reference implementation.

Rolling behavioral windows. Policy replay began at the movement-derived trial start and proceeded chronologically throughout the search. At each update point, the model summarized the participant’s recent behavior using a trailing window. Window duration and update interval were treated as tunable parameters. We evaluated multiple prespecified window–step configurations; the complete set is reported in Appendix A.1.

Feature extraction. For each trailing window, we calculated mean distance to the target, approach ratio, mean movement speed, speed variability, head-yaw entropy, and path tortuosity. Together, these features represented target-directed progress, stopping or variable movement, visual scanning, and indirect navigation, respectively. They were identified as important behavioral indicators of wayfinding difficulty in prior work [33, 35, 52]. Detailed feature definitions are provided in Appendix A.2.

Inferring user states. The window-level features entered a learnable weighted voting procedure that inferred the four states used in Study 1: ON-COURSE, DECISIVE, EXPLORATORY, and STALLED. First, a progress gate assigned ON-COURSE when the participant showed sufficient target-directed progress, based on approach ratio and target proximity. If this gate was not satisfied, the model combined thresholded evidence from speed variability, head-yaw entropy, and path tortuosity to distinguish DECISIVE from EXPLORATORY. Low mean speed combined with high head-yaw entropy provided evidence for STALLED. The state with the highest weighted evidence score was selected; exact ties were resolved in favor of the state associated with the longer patience threshold. The full voting mechanism, including feature thresholds and evidence weights, is reported in Appendix A.3.

Applying patience and triggering assistance. The inferred state activated the same fixed patience values used in Study 1: ON-COURSE suppressed automatic assistance, whereas DECISIVE, EXPLORATORY, and STALLED used patience thresholds of 15, 10, and 5 s, respectively. To prevent a transient inference from a single window from causing rapid, unstable state switching, a newly inferred state had to persist for a tunable confirmation duration before replacing the active state. The model triggered assistance when the active patience threshold had been reached and the target had not yet been retrieved.

Optuna-Based Parameter Optimization. We used Optuna to tune the window duration, update interval, progress-gate thresholds, state-evidence thresholds and weights, and confirmation duration. Within each outer development split, Optuna evaluated 1,000 candidate configurations using a multivariate tree-structured Parzen estimator sampler. Each candidate configuration was assessed through inner cross-validation using only the corresponding outer-development data. Outer-test trials were never used to tune or select the model parameters.

5.3 Calibration Objective

For each candidate configuration, we simulated the automated policy and compared it with the RA reference implementation on each trial. The objective captured both agreement about whether to offer assistance and, when both policies triggered, agreement in trigger time. Let $y_i \in \{0, 1\}$ indicate whether the RAs triggered assistance on trial i , and let $\hat{y}_i$ denote the automated policy’s decision. When both policies triggered, t_i and $\hat{t}_i$ denote the RA and automated trigger times, respectively. We used the following bounded joint trigger–timing loss:

$$\ell_i(\delta) = \begin{cases} 0, & y_i = 0 \text{ and } \hat{y}_i = 0, \\ 1, & y_i \neq \hat{y}_i, \\ \frac{|t_i - \hat{t}_i|}{|t_i - \hat{t}_i| + \delta}, & y_i = 1 \text{ and } \hat{y}_i = 1. \end{cases} \quad (1)$$

A concordant no-trigger decision received zero loss, whereas a trigger-decision mismatch received loss of one. When both policies triggered, loss increased with their timing difference; it equaled 0.5 when the difference was δ and approached one as the difference increased.

To avoid over-weighting participants with more trials or the more prevalent RA-no-trigger class, we first averaged losses within participants and then balanced the two RA reference classes. We defined $c = 0$ as RA no-trigger and $c = 1$ as RA trigger. For participant p , I_{pc} denotes that participant's trials in RA class c , $n_{pc} = |I_{pc}|$, and $\mathcal{P}_c$ contains participants with at least one trial in that class. Thus, the inner average gives each participant's mean loss within a class, the outer average gives each eligible participant equal weight, and the final average gives the two RA classes equal weight:

$$L_c = \frac{1}{|\mathcal{P}_c|} \sum_{p \in \mathcal{P}_c} \frac{1}{n_{pc}} \sum_{i \in I_{pc}} \ell_i(\delta), \quad L = \frac{L_0 + L_1}{2}, \quad (2)$$

We used $\delta = 5$ s as the primary temporal scale and rescored the same held-out predictions at 3 and 10 s for sensitivity analysis. Within each outer optimization split, Optuna selected the configuration that minimized this participant-macro, class-balanced joint loss across the inner cross-validation folds.

5.4 Evaluation Protocol

We evaluated the automated policy under user-dependent and user-independent settings. In the *user-dependent* setting, we used nested five-fold cross-validation to assess performance on held-out trials from participants whose other trials could appear in the outer calibration data. All windows from the same trial were assigned to a single outer fold; thus, no trial appeared in both calibration and test data, although a participant could contribute different trials to both. Within each outer calibration split, Optuna used an inner five-fold cross-validation procedure to select the temporal representation and state-machine parameters.

In the *user-independent* setting, we used nested leave-one-user-out (LOUO) cross-validation. Each outer fold held out all trials from one participant, while all remaining participants provided calibration data. Parameter selection was performed only within those non-held-out participants.

For each held-out trial, we replayed the observed behavioral trajectory from trial start and applied the selected policy at each update point. We recorded whether the policy triggered assistance and, if so, its trigger time. For RA-trigger trials, replay ended at the recorded RA trigger because subsequent behavior could have been altered by the offered assistance. If the automated policy had not triggered by that point, we coded the trial as a reference-discordant no-trigger. For RA-no-trigger trials, replay ended when the participant retrieved the target; a trigger emitted exactly at target retrieval was censored because it could not have preceded task completion. Accordingly, timing results quantify how early the automated policy triggered relative to the RA reference; they cannot establish whether it would have triggered later on an intervention-contaminated trajectory.

We report the primary joint loss, trigger-decision metrics, and timing agreement among jointly triggered trials. We estimated 95% confidence intervals using 2,000 participant-cluster bootstrap resamples.

Table 2. Held-out trigger-decision and paired-timing agreement between the automated policy and the research-assistant (RA) reference.

Metric	User-dependent	User-independent (LOUO)
<i>Trigger-decision agreement (pooled held-out trials)</i>		
Balanced accuracy	0.852	0.847
Precision	0.711	0.719
Recall	0.899	0.878
Specificity	0.804	0.815
F_1	0.794	0.790
<i>Observed paired timing</i>		
Mean AI lead (s)	4.55	4.42
AI lead ≤ 3 s	0.559	0.584
AI lead ≤ 5 s	0.682	0.693
AI lead ≤ 10 s	0.865	0.867

Note. Values are computed from pooled held-out predictions for the same 541 trials in each protocol. Trigger-decision metrics treat an RA trigger as the positive class. Paired-timing metrics are computed only for jointly triggered trials. Because each replay ended at the RA trigger, an automated trigger on a jointly triggered trial could occur only at or before that reference time. AI lead is $t_{RA} - t_{AI}$; thus, "AI lead $\leq k$ s" is the proportion of jointly triggered trials in which the automated policy triggered no more than k s before the RA trigger.

5.5 Results

Table 2 summarizes held-out performance in both evaluation settings.

In user-dependent evaluation, the participant-macro, class-balanced joint loss was 0.338 (95% CI [0.297, 0.382]). At the trigger-decision level, pooled held-out predictions achieved a balanced accuracy of 0.852 (95% CI [0.817, 0.887]), with precision of 0.711, recall of 0.899, and a trigger-decision F_1 score of 0.794. Across 541 held-out trials, the automated policy and the RA reference both triggered assistance in 170 trials (31.4%) and both withheld assistance in 283 trials (52.3%). The automated policy triggered when the RA reference did not in 69 trials (12.8%), whereas the RA reference triggered when the automated policy did not in 19 trials (3.5%).

Among the jointly triggered trials, the automated policy acted an average of 4.55 s before the RA reference. At the primary 5-s scale, 116 of the 189 RA-triggered trials (61.4%) received an automated intervention within 5 s before the RA trigger; this corresponds to 68.2% of jointly triggered trials.

User-independent performance was similar. Across the pooled LOUO held-out trials, the participant-macro, class-balanced joint loss was 0.339 (95% CI [0.298, 0.379]). Trigger-decision performance remained comparable when entire participants were held out, with a balanced accuracy of 0.847 (95% CI [0.812, 0.878]), precision of 0.719, recall of 0.878, and a trigger-decision F_1 score of 0.790. The automated policy and the RA reference both triggered assistance in 166 trials (30.7%) and both withheld assistance in 287 trials (53.0%). The automated policy triggered when the RA reference did not in 65 trials (12.0%), whereas the RA reference triggered when the automated policy did not in 23 trials (4.3%). Among the jointly triggered trials, the automated policy triggered an average of 4.42 s before the RA triggered assistance; 69.3% of them received an automated intervention within 5 s before the RA trigger.

5.6 Study 2 Summary

Study 2 shows that an Optuna-calibrated automated policy can reproduce a substantial portion of the adaptive-patience policy from observable head movement, body movement, target-relative progress, and task context, including for participants not represented in training. Performance was similar in the user-dependent and user-independent settings, suggesting that the automated structure worked robustly on unseen users.

Study 3 evaluates the automated implementation of the AI-triggered Adaptive Patience policy and compares participants' experiences of it during live use with those of the RA-executed implementation.

6 Study 3: Online Evaluation of Automated Adaptive Patience

Study 1 evaluated adaptive patience through a human-mediated implementation: two trained research assistants applied the standardized state machine to observable participant behavior and triggered assistance at the resulting time. Study 2 then automated this policy using a calibrated model. Study 3 closed the loop by examining whether this automated implementation preserved the user experience of the RA-mediated implementation during live interaction, under the same task and environment as in Study 1. We evaluated whether replacing the human executor with the learned trigger changed how participants experienced the timing or usability of the assistant, and how participants interpreted the timing policy (RQ3).

6.1 Participants

We recruited 16 participants through e-mail solicitation. We excluded one participant (P008) because they did not complete the study tasks, leaving 15 participants (8 females, 7 males, and 0 others) for analysis. None had participated in Study 1. Exact ages were available for 13 participants (22–29 years; $M = 24.77$, $SD = 2.59$); the remaining two reported being 23 years of age or older. Participants self-identified as East Asian ($n = 11$), Southeast Asian ($n = 3$), or Black or of African heritage ($n = 1$). Participants were assigned IDs P001–P016, with P008 excluded. Each study session lasted around 45 minutes and participants were compensated with \$15.

6.2 Conditions and Procedure

Study 3 used a within-subjects design with two versions of adaptive patience:

- **Adaptive Patience (RA-triggered) (C3).** This condition reproduced the human-mediated adaptive condition from Study 1. Two trained RAs monitored the participant's observable behavior and executed the standardized adaptive-patience policy.
- **Adaptive Patience (AI-triggered) (C4).** The optimized model from Study 2 executed the trigger policy online and automatically determined when the assistant appeared. The model was applied in Unreal Engine in real time. All the parameters including window duration, step, thresholds, and voting weights were based on the trained outcome of Study 2, using the 69 participants' data.

After providing consent, participants completed training on locomotion, object manipulation, and interaction with the assistant. They then completed two 3-minute condition blocks, each comprising up to nine sequential ingredient-search trials, in randomized order across Conditions C3 and C4. Following each block, participants and the interviewer reviewed the participant's screen recording on a trial-by-trial basis. For every trial in which the assistant did not offer help, participants were asked whether they had needed assistance. For every trial in which the assistant offered help, participants completed the four-item intervention-timing survey and a brief video-based interview. After completing the video review for each condition, participants completed the System Usability Scale. After both condition blocks and their associated video-review sessions, participants completed a final semi-structured interview. Sessions lasted approximately 45 minutes. The study was approved by the institutional review board at [redacted for peer review].

6.3 Measures and Analysis

Measures. After each condition, the participant and interviewer reviewed the participant’s screen recording for each trial separately. For *every* trial in which the assistant did not help, we interviewed the participant about whether they had needed help.

For *every* trial in which the assistant offered help, participants completed a survey with four intervention-timing ratings, and then went through a short video-based interview.

For the intervention-timing ratings, participants rated four statements on a five-point scale about disruption (Q1), intervention at the wrong time (Q2), timing satisfaction (Q3) and perceived helpfulness (Q4). Q1-Q3 were selected because they were intervention-timing-related metrics and were rated as significant or descriptively significant between fixed patience and adaptive patience groups from Study 1. Q4 was added to measure the general helpfulness of the timing of the assistance. Lower ratings therefore indicate better outcomes for Q1 and Q2, whereas higher ratings indicate better outcomes for Q3 and Q4.

For the video-based interview, participants described what they were about to do, whether they needed help, what they believed caused the assistant to appear, and whether they would have preferred the intervention earlier, later, at the observed time, or not at all. These questions were designed to contextualize individual trigger decisions and learn from user feedback.

After the video review of each condition, participants completed the 10-item System Usability Scale (SUS). Responses were converted to the standard 0–100 SUS score. After both conditions, participants completed a final semi-structured interview comparing the two versions’ ability to recognize when help was needed. During this final interview, we also asked participants to hypothetically compare AI-triggered assistance with self-triggered assistance and to identify settings in which AI-triggered help would or would not be appropriate.

The completed video survey, video interview script and final interview script are available in Appendix B.

Intervention-Timing Ratings and System Usability Analysis. The four intervention-timing items were completed after each assistance event; therefore, participants could contribute different numbers of event-level ratings across conditions. To retain the participant as the inferential unit, we first averaged valid ratings within each participant, condition, and question. We then compared the resulting RA-triggered and AI-triggered participant means using two-sided paired Wilcoxon signed-rank tests. We controlled the false discovery rate across the four question-level comparisons using the Benjamini–Hochberg procedure. For each question, we also report the matched rank-biserial correlation and the mean paired difference (AI-triggered minus RA-triggered) with a participant-level percentile bootstrap 95% confidence interval (10,000 resamples).

For SUS, we compared paired 0–100 scores using a two-sided Wilcoxon signed-rank test. We additionally report the mean paired difference (AI-triggered minus RA-triggered) with a participant-level percentile bootstrap 95% confidence interval and the Pearson correlation between matched RA-triggered and AI-triggered SUS scores, with a Fisher-transformed 95% confidence interval. Because SUS constituted one prespecified comparison, no multiple-comparison correction was applied. All hypothesis tests were two-sided with $\alpha = 0.05$.

Interview Analysis. We used codebook thematic analysis [15] to analyze event-level video reviews (RA-triggered: 47 assistance trials, 69 no-assistance trials; AI-triggered: 44 assistance trials, 79 no-assistance trials) and final interviews from all 15 Study 3 participants. We aimed to explain how participants interpreted intervention need and timing. Two researchers independently applied a shared deductive codebook covering perceived need, timing, RA–AI differences,

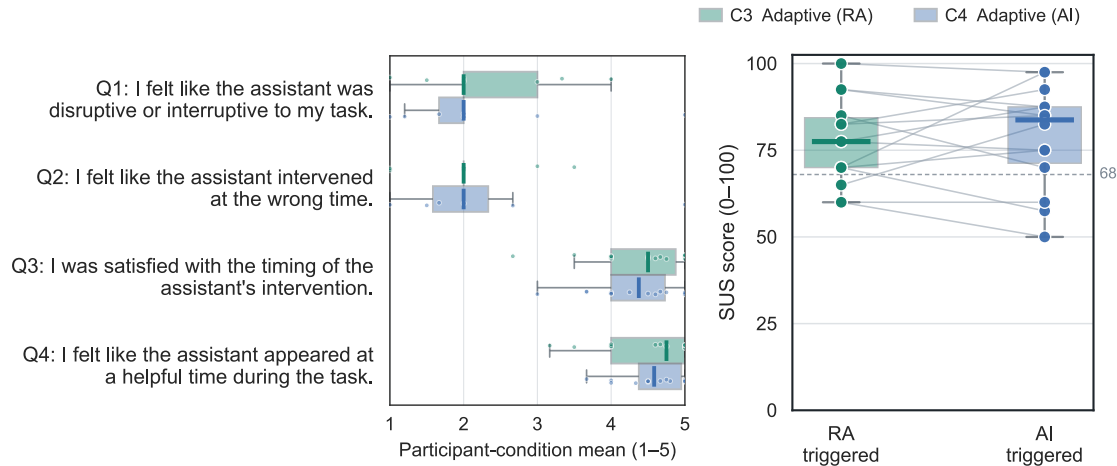


Fig. 5. Participant-level timing-related survey ratings and paired System Usability Scale (SUS) scores. Left: dots are participant-condition means across intervention events; boxes show the interquartile range and median. Lower scores indicate more favorable ratings for Q1–Q2, whereas higher scores are more favorable for Q3–Q4. Right: lines connect each participant’s paired SUS scores ($n = 14$); the dashed line marks the conventional SUS benchmark of 68.

understandability, and proactive versus self-triggered assistance, while adding inductive codes including elapsed search time, hesitation, repeated scanning, wrong selections, movement away from the target and missed targets. The researchers then compared their independently added codes, discussed differences in interpretation, and consolidated overlapping codes into a shared analytic codebook. Examining relationships among these codes showed that individual cues were repeatedly interpreted through participants’ task progress and the remaining value of continued activity. Candidate themes were reviewed against the dataset, including accounts that challenged their scope, yielding three themes.

6.4 Results

6.4.1 Intervention-Timing Ratings. Figure 5 left panel provides a participant-level view of the timing-related ratings. The two conditions showed substantially overlapping distributions across all four items. Under both implementations, ratings were generally low for disruption and wrong timing ($M = 1.81$ – 2.36) and high for timing satisfaction and helpful timing ($M = 4.29$ – 4.55). At the same time, the plotted distributions reveal individual variability: although favorable ratings predominated, some participant-condition means fell at the less favorable end of the scale, particularly for disruption and wrong timing.

At the group level, the estimated mean paired differences in participant-level timing ratings (AI-triggered minus RA-triggered) were small, ranging from -0.03 to 0.13 points on the five-point scale. All 95% bootstrap confidence intervals included zero, and none of the paired Wilcoxon tests was significant after BH-FDR correction (all $q_{BH} = 1.00$; $r_{tb} = -.13$ to $.06$). Thus, we found no evidence that automating the trigger systematically made interventions feel more disruptive, more poorly timed, less satisfying, or less helpful.

6.4.2 System Usability. The paired SUS scores also showed a broad but largely overlapping range across the two implementations (Figure 5, right). Among the 14 participants with paired scores, reported usability was favorable for

both the RA-triggered condition ($M = 77.50$, $SD = 12.40$) and the AI-triggered condition ($M = 78.75$, $SD = 14.86$). The mean paired difference was 1.25 SUS points (AI-triggered minus RA-triggered; 95% percentile-bootstrap CI $[-4.29, 7.68]$). The paired Wilcoxon test found no detectable difference between these conditions ($W = 43.50$, $p = .889$, $r_{tb} = .04$).

Matched SUS scores were positively correlated across conditions ($r = .63$, 95% Fisher CI $[.15, .87]$, $p = .016$): participants who rated the RA-triggered implementation as more usable also generally rated the AI-triggered implementation as more usable, suggesting consistency in participants' relative usability judgments across implementations. Both condition means exceeded the conventional SUS benchmark of 68 [13, 31], indicating generally favorable reported usability in this sample.

6.4.3 Interview Findings. We identified three themes from participants' event-level reviews and final interviews. Participants evaluated whether assistance was timely by considering their current state, the time already invested, the expected value of continuing to search independently, and their preferences for autonomy and control.

Timing fit reflected sustained, unproductive search. Participants generally understood appropriate intervention as a judgment based on both user state and accumulated time, rather than as a response to an isolated cue. For example, P006 attributed the intervention to having "probably been looking for quite a while without finding it." P007 similarly explained that it appeared "probably because of the time I had spent and because I was wandering around the wrong area." P001 endorsed this logic of inferring need from a sustained state: "But if it was trying to guess when I started to feel confused, I think that timing was quite right." P016 likewise reasoned that the system may have appeared because they "looked to the right and the left and still couldn't find it," and they therefore thought that the AI assistant's timing was "quite right."

Thus, from participants' point of view, repeated scanning, slow movement, or wrong selections did not directly imply system intervention. Instead, participants regarded these cues as meaningful when they persisted without producing expected progress. These accounts suggest that participants found a state-sensitive waiting policy intelligible when it intervened after a sustained pattern of unproductive search.

Participants also distinguished between eventually needing help and needing help *now*. Independent search retained value when users still had a plausible action worth trying. In an event in which the AI did not trigger, P016 said that they did not yet need assistance: "Not really, because I was trying to figure it out." Even when behavior appeared highly exploratory, participants could still view the remaining value of continued independent search as high; in those cases, intervention risked interrupting an ongoing strategy. P006 explained, "I did need help, but I had not finished looking—I had not finished trying things yet, although I was almost done." In contrast, shorter waiting was appropriate when further action would only repeat an unsuccessful strategy or when users had no viable next step. P016 thought the intervention while searching for tuna was well timed because "it showed up at the right time because I had already tried to find it but couldn't." P014 recalled, "I was just standing there because I had absolutely no idea what it was," and thought that the system "could have appeared earlier." Therefore, timing fit depended not only on whether users appeared hesitant or stalled, but also on the remaining expected value of autonomous search in that state.

Patience preferences and search strategies were personal and context-dependent. Participants differed in how much time they wanted the system to allow for independent search. Some preferred longer waiting. P015 noted that the system may have seen that they had "already looked at many objects," but felt that they "might keep looking for a little while and perhaps find it"; accordingly, the assistant "could have appeared a bit later." Others wanted earlier intervention. P001 stated that, if given a help button, "I would definitely have pressed it earlier." For several participants, this directional

preference recurred across multiple reviews: when they explicitly evaluated timing, P001 and P009 repeatedly described assistance as too late, whereas P015 repeatedly described it as too early. Therefore, beyond user state and time, clear individual preference emerged as a recurring tendency that should inform help trigger time, potentially by adding or removing offsets from the patience threshold based on personal preference.

This variation partly reflected users' own search strategies, whose value was not always visible in their behavior. P009 regarded a longer initial search as a way of learning the environment: "I could find things much faster later because the first time I spent longer looking around and got to know it." Accordingly, when the system intervened during exploration, P009 felt that "when I wanted to explore, it would not let me explore; it just popped up." P013 similarly wanted time to establish an overview of the space: "I need some time to look at the whole kitchen first before the assistant appears." Personal experience also shaped seemingly indirect actions. P001 explained that they checked cabinets because they normally stored soy sauce in a cabinet at home. These accounts show that behaviors such as moving away from the target, repeatedly scanning, or pausing may also reflect a user's independent strategy not tied to stalled stages or help needs. Systems that infer intent from observable behavior should therefore preserve uncertainty and allow users to calibrate or defer assistance.

Proactivity reduced help-seeking burden, but users weighed its benefits against agency and context. Proactive assistance reduced the burden of recognizing a need for help and explicitly requesting it. P006 noted that, when "reluctant to give up", they might continue searching and waste time; if the system appeared on its own, "it would be more efficient." This convenience was especially valuable when users *had difficulty asking* for help or did not immediately realize they needed help. P016 suggested that "[for] someone who doesn't know how to operate or control the controller, it may be easier for them to get help." P001 likewise noted that proactivity could be useful if "my hands were not free, or I couldn't really press anything" and that they would prefer proactive assistance to a natural-language trigger because "speaking is tiring"; P016 extended this benefit to everyday situations, such as receiving a reminder that water had been boiling for too long and that the heat needed to be turned off.

While proactive assistance can reduce help-seeking effort, some participants also noted that a sense of control matters. P006 argued that "my own timing would definitely be 100% accurate," while P013 said, "I would just prefer something that I can control." P012 characterized self-triggered help as "more flexible" and offering "more autonomy," and P015 worried that direct assistance could make users lose "the ability to explore on [their] own." Thus, participants valued proactivity because it reduced the friction of obtaining help when needed, but a self-trigger option may help improve trust by providing the sense of control for some participants. Proactive systems should therefore combine automatic intervention with ways to retain, defer, or override assistance, supporting users during sustained, unproductive search without taking away meaningful opportunities for independent action.

6.5 Study 3 Summary

Study 3 evaluated whether the learned policy could deliver adaptive patience during live interaction without degrading the experience of the RA-mediated implementation. Across 15 participants, we found no evidence of systematic differences in perceived disruption, timing appropriateness, timing satisfaction, helpfulness, or usability; both implementations received generally favorable system usability ratings above the conventional SUS benchmark. Interviews also showed that participants judged assistance as appropriate when unproductive search persisted, but their preferred waiting time depended on their search strategy, context, and desire for autonomy, which we discuss further in the Discussion.

7 Discussion

Across the three studies, our findings suggest that timing proactive assistance can be a temporally extended, contextual judgment that considers a user's current state and how long the difficulty has persisted. This judgment balances the opportunity for potentially valuable independent action against the cost of leaving a possible need for assistance unaddressed. Adaptive Patience operationalizes this balance by mapping each interpretable user state to a corresponding waiting period, while allowing meaningful target-directed progress to defer intervention.

7.1 Adaptive Patience Treats Timing as a Sustained, Contextual Judgment

Our findings position "patience" as a mechanism for allocating time for independent action. In this setting, Study 1 provides experimental evidence that adapting the waiting period to a user's state improved several aspects of the proactive-assistance experience relative to a state-insensitive fixed wait. Compared with Fixed Patience, Adaptive Patience was perceived as more partner-like, less disruptive, and less likely to intervene at the wrong time. Timing satisfaction was also descriptively higher. Relative to Self-Triggered assistance, Adaptive Patience was perceived as more partner-like, but also as more disruptive and less controllable. The two conditions did not differ detectably in perceived mistiming or timing satisfaction. These tradeoffs indicate that proactive timing should not be understood as replacing user control. Rather, its role is to offer assistance when continued waiting may no longer be wanted, while retaining room for users to pursue viable strategies and to request, defer, or decline help.

Study 3 provides further insight into how participants evaluated this judgment during live interaction. Both RA- and AI-triggered Adaptive Patience received generally favorable timing ratings and usability scores, with no detected group-level differences between implementations. This result supports the feasibility of bringing the policy into live interaction. In interviews, participants evaluated timing in relation to an unfolding combination of their current state, the time already invested, whether they were making target-directed progress, and the expected value of continuing to search independently. They generally wanted more waiting when they were decisively pursuing a coherent plan or actively learning the environment, and earlier assistance when their search had stalled.

This account clarifies the role of the policy components. The ON-COURSE progress gate and the longer patience assigned to DECISIVE encode a design commitment to preserve productive, self-directed exploration before providing highly directive help. Similar protections may be relevant in other proactive systems, such as game hints, tutoring tools, or task assistants that can rapidly reveal a solution or complete part of a task.

7.2 Human-Specified Timing Policies Can Be Automated

Study 2 shows that a human-specified timing policy can be translated into an executable implementation. Using behavioral and environmental signals available to the system, the Optuna-calibrated policy achieved substantial agreement with the RA reference in held-out trials and participants. This result indicates that the state-patience-progress structure is sufficiently explicit to serve as a computational target: its parameters can be calibrated from observed data and then executed during interaction.

Policy-derived supervision may be particularly useful for proactive assistance because it aligns the learning target with an explicit timing rationale. Self-triggered requests remain valuable signals of users' self-perceived need for support. However, the decision to request help can also depend on factors that are not directly observable to a system. Thus, the timing policy operationalizes an explicit mapping from observable behavior and task context to a timing decision, providing an inspectable and deployment-relevant target for calibration.

7.3 Patience Should Account for Strategy Beyond the Current Trial

Observable behavior remains useful evidence about how a user is currently engaging with a search. Within a trial, looking around, revisiting areas, repeatedly scanning, or moving slowly may signal unproductive search, but may also reflect orientation or information gathering. Adaptive Patience addresses part of this ambiguity by offering help in response to a user state that persists over time, rather than to a single observed behavior. Our interviews, however, point to a second, longer-horizon source of ambiguity: the same local behavior can also be part of a strategy that spans multiple trials.

Participants described deliberately using early searches to learn the environment’s layout, even when doing so slowed immediate target acquisition. In such cases, slow movement, revisiting locations, or unsuccessful selections may represent an investment in information that makes later searches easier. Accordingly, an early search in an unfamiliar environment may require more exploration time than a later search after familiarity has been established, even when the momentary behavior appears similar.

This context can also change within a person, rather than reflecting a fixed individual preference: a user may prefer more patience while learning an environment and less once their goal shifts toward efficient completion. Future systems could combine current user state and progress with cross-trial information, such as spatial coverage, previously inspected regions, prior assistance, and evidence of environmental familiarity, to adjust state-specific patience or identify an episode-level orientation phase. When users’ broader goals remain uncertain, a lightweight check-in may be preferable to immediate guidance.

7.4 Proactivity Should Remain User-Controllable and Personalized

Finally, our findings suggest that proactive and self-triggered assistance should not be treated as mutually exclusive modes. Study 1 highlights the importance of retaining a user-controlled route to support through self-triggered requests, whereas the Study 3 interviews suggest that proactive assistance may be particularly useful when a task is cognitively demanding, the environment or interaction is unfamiliar, or users cannot reliably notice and articulate their need for help. In such situations, asking for help can itself impose attention and coordination demands. Proactivity may reduce that burden, but it can also make the loss of control more apparent.

The appropriate future design goal is therefore not a single globally optimal patience threshold, but a personalized, state-dependent patience function that depends on both user and current state. In practice, this may be better understood as a range of acceptable waiting times than as one universally correct instant for intervention. One person may prefer a long waiting period while deliberately exploring but a short one when stalled; another may prefer more or less system initiative overall. Future systems could retain an always-available request channel while allowing users to dismiss, defer, or temporarily disable proactive offers. When help becomes eligible, the system could first issue a lightweight check-in and provide full guidance only after acceptance. Over time, users’ acceptance, dismissal, and deferral behavior could update state-specific patience estimates. Such hybrid designs preserve the potential benefit of reducing help-seeking burden without requiring users to surrender control over when and how assistance enters the interaction.

7.5 Limitations and Future Work

We draw our conclusions from the current setting. We evaluated the policy in a VR kitchen, where assistance directly revealed a target’s location; its state definitions and patience thresholds may not transfer unchanged to other tasks or less directive forms of assistance. Although we argue that Adaptive Patience offers a useful approach to timing

proactive AI in open-ended tasks, deployment requires task-specific decisions about which user states are meaningful and how long the system should wait in each state.

Moreover, Study 2 evaluated automation against RA trigger times. These references may include real-time observation errors, ambiguity in interpreting a user's state, and timing variability. Future work could establish a stronger reference set by having multiple reviewers inspect recorded trajectories, videos, and environmental context offline. Rather than annotating only a single trigger instant, reviewers could independently label the current user state, meaningful target-directed progress, and an acceptable intervention window. This would separate disagreement about state interpretation from reaction-time noise and provide a more auditable basis for calibrating and evaluating automated policies.

Finally, the automated policy relied on a limited, literature-informed set of movement, distance, and speed features. These features capture only part of the richer, evolving contextual judgments an RA can make, and the RA decisions used as reference labels are themselves imperfect. Future work should test richer contextual signals, independently validated policy labels, longitudinal personalization, and a wider range of task domains. Such studies can help establish when adaptive patience improves not only perceived intervention timing, but also users' longer-term autonomy and task experience.

8 Conclusion

This work examines how proactive AI can decide when to assist by dynamically adjusting the time it waits for users' independent exploration based on their current task-progress state. We introduce *Adaptive Patience*, an interpretable, user-state-sensitive policy for timing proactive assistance. In an embodied spatial-search task, the policy maps observable search behavior to state-specific patience thresholds, allowing continued independent exploration when users appear to be making progress and shortening the waiting period when difficulty persists.

Across three studies, we evaluated the policy, its automation, and its live implementation. In a within-subject Wizard-of-Oz study ($N = 69$), compared with Fixed Patience, Adaptive Patience was perceived as more partner-like, less disruptive, and less likely to intervene at the wrong time, without detectable condition differences in objective task performance. Compared with Self-Triggered assistance, Adaptive Patience was also perceived as more partner-like, but as more disruptive and less controllable; perceived wrong-time intervention and timing satisfaction did not differ detectably between these two conditions. These findings locate the benefit of adaptation of waiting patience based on observable user state, while also highlighting the tradeoff between proactive initiative and direct user control.

We then used Optuna-based parameter optimization to calibrate an automated, policy-constrained implementation from observable behavioral signals. The automated policy achieved a balanced accuracy of 0.852 in the user-dependent setting and 0.847 in the user-independent setting.

Finally, in a live within-subject study ($N = 15$), AI-triggered and RA-triggered Adaptive Patience showed no evidence of systematic differences in perceived disruption, timing appropriateness, timing satisfaction, helpfulness, or usability. Both implementations received favorable usability ratings (AI-triggered: $M = 78.75$; RA-triggered: $M = 77.50$). Interviews further showed that participants judged assistance as appropriate after sustained, unproductive search, while their preferred waiting time depended on progress, search strategy, context, and desired autonomy.

Together, these findings position proactive timing as a temporally extended decision: assistance should reflect how a user is engaging with a task, how long that state persists, and the value of preserving independent exploration. Adaptive Patience offers an interpretable and practical structure for designing such support.

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Table 3. Window-level behavioral features used for user-state inference.

Feature	Definition within the trailing window
Mean distance to target	Mean Euclidean distance between the participant's position and the current target.
Approach ratio	Proportion of valid movement intervals in which the participant moved toward the target.
Mean movement speed	Mean locomotor speed across valid movement samples.
Speed variability	Coefficient of variation of movement speed, capturing changes between sustained and intermittent movement.
Head-yaw entropy	Entropy of the participant's head-yaw distribution, with larger values indicating more dispersed visual scanning.
Path tortuosity	Calculated by fractal dimension; indirectness of movement relative to target-directed travel, with larger values indicating a less direct path.

A Temporal Windows, Feature Construction, and State Voting

This appendix specifies the temporal representations, window-level features, and weighted voting mechanism used to infer user states in the automated adaptive-patience policy.

A.1 Temporal Representations

Beginning at the movement-derived trial start, the policy evaluated the participant's behavior using successive trailing sliding windows. Each temporal representation was defined by a window duration w and a sliding-window step s : w determined how much recent behavior informed a state estimate, whereas s determined the temporal distance between consecutive windows and, consequently, how often the policy updated that estimate. We prespecified 11 (w, s) configurations (in seconds): (2, 0.5), (2, 1), (3, 0.5), (3, 1), (3, 1.5), (4, 0.5), (4, 1), (4, 2), (5, 0.5), (5, 1), and (5, 2).

A.2 Window-Level Features

For every trailing window, we derived the six behavioral features in Table 3. Each feature was calculated using only observations available at that update point.

A.3 Weighted Voting for User States

At each update point, the model applied the following state-inference procedure. Let $\bar{d}$, a , $\bar{v}$, c , h , and q denote mean target distance, approach ratio, mean movement speed, speed variability, head-yaw entropy, and path tortuosity in the current window, respectively.

State inference at each update point**Input:** current window features $(\bar{d}, a, \bar{v}, c, h, q)$.

- (1) **If** the target has been retrieved, stop policy replay; do not issue assistance.
- (2) **Else if** $a \geq \theta_a$, assign ON-COURSE.
- (3) **Else if** both $\bar{d} \leq \theta_d$ and $a \geq \theta_{\text{near}}$, assign ON-COURSE.
- (4) **Else**, initialize the three help-eligible state scores to zero: $S_{\text{decisive}} = S_{\text{exploratory}} = S_{\text{stalled}} = 0$.
 - (a) **If** $c \geq \theta_c$, add w_c^E to $S_{\text{exploratory}}$; otherwise, add w_c^D to S_{decisive} .
 - (b) **If** $h \geq \theta_h$, add w_h^E to $S_{\text{exploratory}}$; otherwise, add w_h^D to S_{decisive} .
 - (c) **If** $q \geq \theta_q$, add w_q^E to $S_{\text{exploratory}}$; otherwise, add w_q^D to S_{decisive} .
 - (d) **If** both $\bar{v} \leq \theta_v$ and $h \geq \theta_h$, add w_s to S_{stalled} .
- (5) Assign the state with the highest score. Resolve exact ties in favor of the state associated with the longer patience threshold.

Thus, target-directed movement first prevents unnecessary intervention through the ON-COURSE gate. When that gate is not satisfied, variable movement, broad visual scanning, and indirect paths cast weighted votes toward EXPLORATORY; lower values on these features vote toward DECISIVE; and low movement speed combined with broad scanning adds evidence for STALLED.

All thresholds and evidence weights were selected using only the corresponding inner cross-validation data. Because nested cross-validation selected a separate parameter configuration for each outer training split, no single set of threshold or weight values underlies all held-out results.

After state selection, the policy applied the fixed patience mapping from Study 1: ON-COURSE suppressed automatic assistance, whereas DECISIVE, EXPLORATORY, and STALLED used patience thresholds of 15, 10, and 5 s, respectively. A newly inferred state had to persist for the selected confirmation duration before becoming active. Changing the active state updated the applicable patience threshold but did not reset elapsed active-search time. The policy emitted assistance at the first update point for which the active patience threshold had been reached and the target had not yet been retrieved.

B Study 3 Usability-Test Interview Protocol

This appendix presents the interview protocol used in Study 3. After each condition, participants reviewed a screen recording of their interaction with the assistant. The interviewer asked the following questions for each relevant intervention event and for segments in which the assistant did not intervene. After participants completed both conditions, they took part in a post-study interview.

B.1 Post-Condition Video Review

Participants received the following instructions:

After completing each condition, we will review your screen recording together. For the following statements, please indicate your agreement on a 5-point scale, where 1 indicates *Strongly Disagree* and 5 indicates *Strongly Agree*.

B.1.1 When the Assistant Intervened. For each instance in which the assistant appeared, participants rated the following statements:

- (1) The assistant was disruptive or interruptive to my task.
- (2) The assistant intervened at the wrong time.

- 1405 (3) I was satisfied with the timing of the assistant's intervention.
 1406 (4) The assistant appeared at a helpful time during the task.
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1408 The interviewer then asked the following open-ended questions:

- 1409 (1) **Task context:** "At the exact moment the assistant appeared, what were you about to do?"
 1410 *Interviewer note: Probe for participants' specific actions at that moment rather than only their general thoughts.*
 1411 (2) **Perceived need:** "At that moment, did you feel that you needed help? Why or why not?"
 1412 (3) **Perceived trigger:** "What do you think caused the assistant to appear at that moment?"
 1413 *Optional prompt: "For example, do you think it responded to your speed, hesitation, or the amount of time you spent*
 1414 *on the task?"*
 1415 (4) **Preferred timing:** "Would you have preferred the assistant to appear at that moment, earlier, later, or not at
 1416 all?"
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1420 *B.1.2 When the Assistant Did Not Intervene.* For each segment in which the assistant remained hidden, the interviewer
 1421 asked:

- 1422 (1) "During this specific segment, the assistant did not appear. Did you actually need its help at that time?"
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1425 B.2 Post-Study Interview

1426 After completing and reviewing both conditions, participants answered the following questions:

- 1427 (1) "What was the biggest difference between the two versions in their ability to 'read your need'—that is, to
 1428 recognize whether you actually needed help?"
 1429 (2) "Which version was better at recognizing when you needed help? What specific experiences led you to that
 1430 conclusion?"
 1431 (3) "In other settings, help is provided only when you actively request it, meaning that you decide when the assistant
 1432 appears. Compared with self-triggered assistance, what were the main benefits or drawbacks of AI-triggered
 1433 assistance?"
 1434 (4) "In what situations would AI-triggered assistance be useful or appropriate, and why?"
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